

PROJECT EXPO 2026

Smart IV Drip Monitoring Using IOT

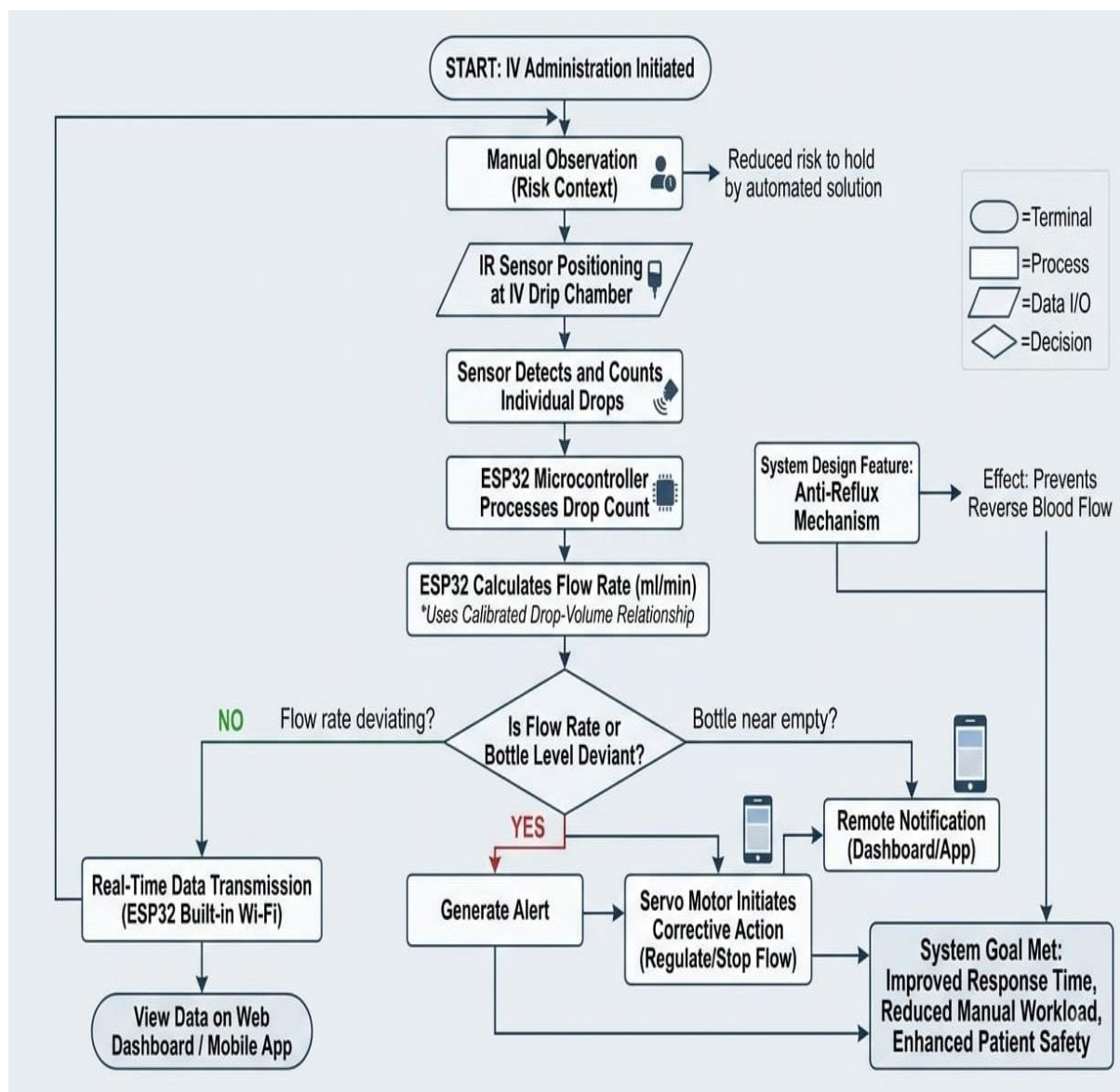
TEAM MEMBERS:

M.DHASLIMA BANU

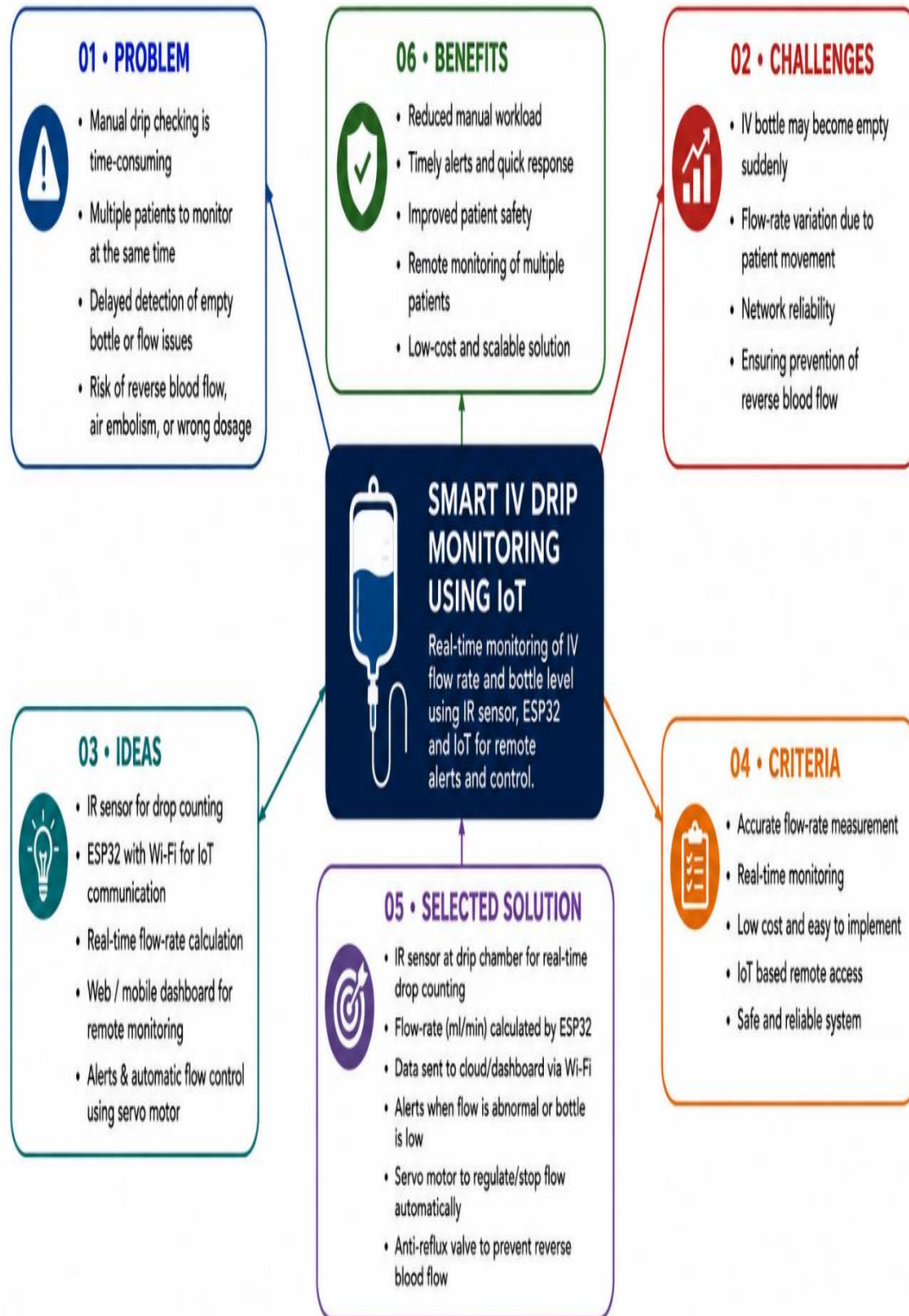
P. GANGASHREE

FLOWCHART:

SMART IV DRIP MONITORING SYSTEM (IOT-Enabled Healthcare Solution)



BRAINSTROMING REPORT:



IDEA PRIORITIZATION MATRIX:

Idea	Feasibility	Innovation	Cost	User Benefit	Priority
 IR Drop Monitoring	High	High	Low	Very High	Very High
 Flow-Rate Calculation	High	High	Low	Very High	Very High
 IoT Dashboard	High	High	Low	High	High
 Servo-Based Control	Medium	High	Medium	High	High
 Anti-Reflux Safety	High	Medium	Low	Very High	High



Key Findings:

- **IR-based drop sensing** was selected as the primary sensing method because it directly measures individual IV drops.
- **Flow-rate calculation** converts drop information into a useful ml/min monitoring value.
- **IoT communication** enables remote visibility of IV status and abnormal conditions.
- **Servo control and anti-reflux mechanisms** provide additional safety-oriented functionality.

EMPATHY MAP:

Primary Persona: Hospital Nurse



USER PERSONA:

Name :Priya Sharma

Age :32 Years

Occupation :Staff Nurse

Experience :6 Years

Responsibilities:

- Monitor IV fluid flow and bottle status.
- Respond to abnormal flow or empty-bottle alerts.
- Maintain patient safety during IV therapy.

User Persona – Goals and Challenges:

Goals:

- Reduce frequent manual IV inspection.
- Improve patient safety.
- Receive timely alerts for abnormal flow.
- Save time during nursing rounds.

Challenges:

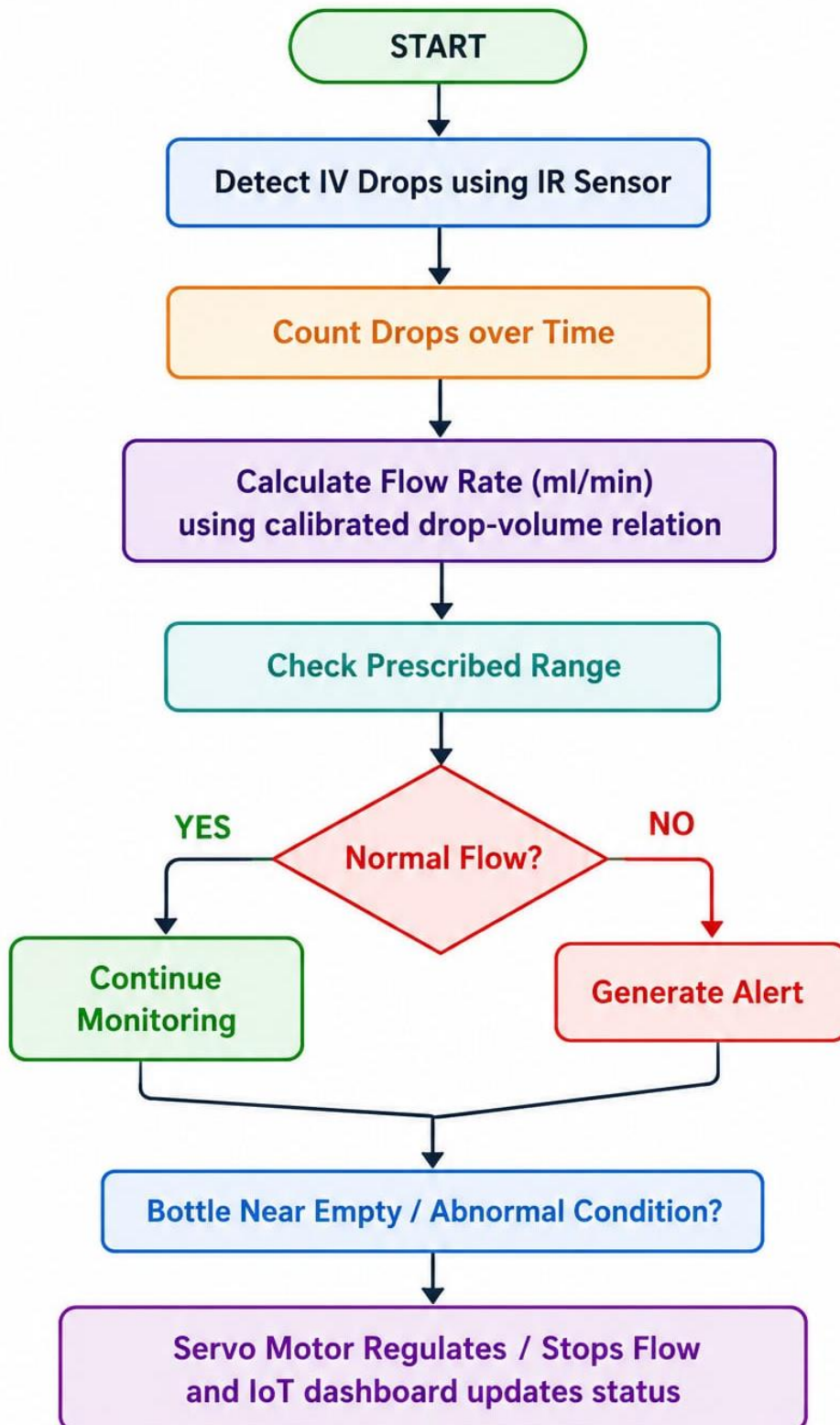
- Monitoring multiple IV patients simultaneously.
- Delayed identification of an empty or near-empty IV bottle.
- Flow-rate variations may go unnoticed.
- Increased workload during busy shifts.

PROPOSED SOLUTION:

Smart IV Drip Monitoring using IoT



FUNCTIONAL REQUIRMENTS:



NON-FUNCTIONAL REQUIRMENTS:

	Accuracy	Provide reliable drop-count and flow-rate measurements.
	Reliability	Operate continuously with minimal sensor or software failure.
	Responsiveness	Detect abnormal flow and update alerts quickly.
	Usability	Simple dashboard for quick monitoring and decisions.
	Safety	Use regulated low-voltage power and safe isolation.
	Connectivity	Maintain stable Wi-Fi communication where available.
	Scalability	Support additional IV channels in future.
	Maintainability	Allow easy calibration, sensor replacement and testing.
	Low Power	Use efficient power operation for continuous monitoring.
	Data Integrity	Maintain consistent monitoring data and alert status.

PROJECT PLANNING:

ACTIVITIES	AUG W1	AUG W2	AUG W3	AUG W4	SEP W1	SEP W2
Requirement Analysis	●	●				
Component Procurement	●	●				
Hardware Assembly		●	●			
IR Sensor Interfacing			●	●		
ESP32 Programming			●	●		
Flow-Rate Calibration			●	●	●	
Servo Control Integration				●	●	
IoT Dashboard Development				●	●	
Alert Testing					●	
Anti-Reflux Safety Check					●	●
System Integration					●	●
Testing & Calibration					●	●
Documentation						●

Legend: ● = Planned Activity

RISK ANALYSIS:

Smart IV Drip Monitoring System using IoT

ID	RISK	IMPACT	LIKELIHOOD	MITIGATION	CONTINGENCY
R1	IR sensor miscount	Wrong flow-rate estimate	Medium	- Calibrate sensor - Shield from ambient light	Recalibrate, replace sensor
R2	Wi-Fi communication loss	Remote alert may be delayed	Medium	- Auto-reconnect - Check signal strength	Store status locally (in device's memory)
R3	Servo malfunction	Flow may not stop/regulate	Low	- Use servo with mechanical limit/timing - Regular testing	Manual clamping of IV tube
R4	Power failure	Monitoring stops	Medium	- Regulated supply - Backup power (battery/UPS)	Restart safely and check system
R5	Software error	Incorrect status or alerts	Low	Proper testing and watchdog restart logic	Manual reset and update

Risk Level:

Low

= Unlikely to occur

Medium

= May occur occasionally

High

= Likely to occur

